

DIRECTED ENERGY WEAPONS:

Particle Beam Weapons



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Laser is no longer confined to premises of prominent research centres like Bell laboratories, Hughes Research Laboratories and major academic institutes like Columbia University, USA, as it was in its early stages of development and evolution. In the last five decades, after Theodore Maiman demonstrated the first laser in May 1960 at Hughes Research Laboratories, there has been explosive growth in industrial, medical, scientific and military applications of lasers. Application areas are continuing to grow with every passing day.

Lasers have been used in various military applications since the early days of development that followed their invention. There has been large-scale proliferation of lasers and optronic devices and systems for applications like range finding, target designation, target acquisition and tracking, precision-guided munitions and so on during 1970s and 1980s.

These devices continue to improve in performance and find increased accept-

ance and usage in contemporary battlefield weaponry. Technological advances in optics, optoelectronics and electronics leading to more rugged, reliable, compact and efficient laser devices are largely responsible for making these indispensable in modern warfare.

Last one decade or so has also seen emergence of a new class of weapons known as directed energy weapons (DEWs) leading to enhanced global interest from scientists and engineers in DEWs' development. Lasers, high-power microwaves and high-energy particle beams have been exploited for DEW development. These weapons, with the exception of particle beam weapons (PBWs) and laser-induced plasma channel (LIPC) weapons, generate streams of electromagnetic energy that can be precisely directed over long distances to disable or destroy intended targets.

After decades of research and development, DEWs are now becoming an operational reality. This has been possible due to their unique characteristics that potentially enable new concepts of military operation and also because there has been considerable progress over the past two decades in developing relevant technologies such as power sources, beam-control concepts and pointing and tracking techniques. For these applications, lethal energy from a high-power laser or a source of high-power microwaves or high-energy particle beam is delivered to the targets for causing either neutralisation of electro-optic sensors onboard the target platform or structural damage to the target itself.

DEWs

A DEW system, with the exception of LIPC weapons, primarily uses directed energy in the form of concentrated beam of

Fig. 1: Archimedes death ray concept

